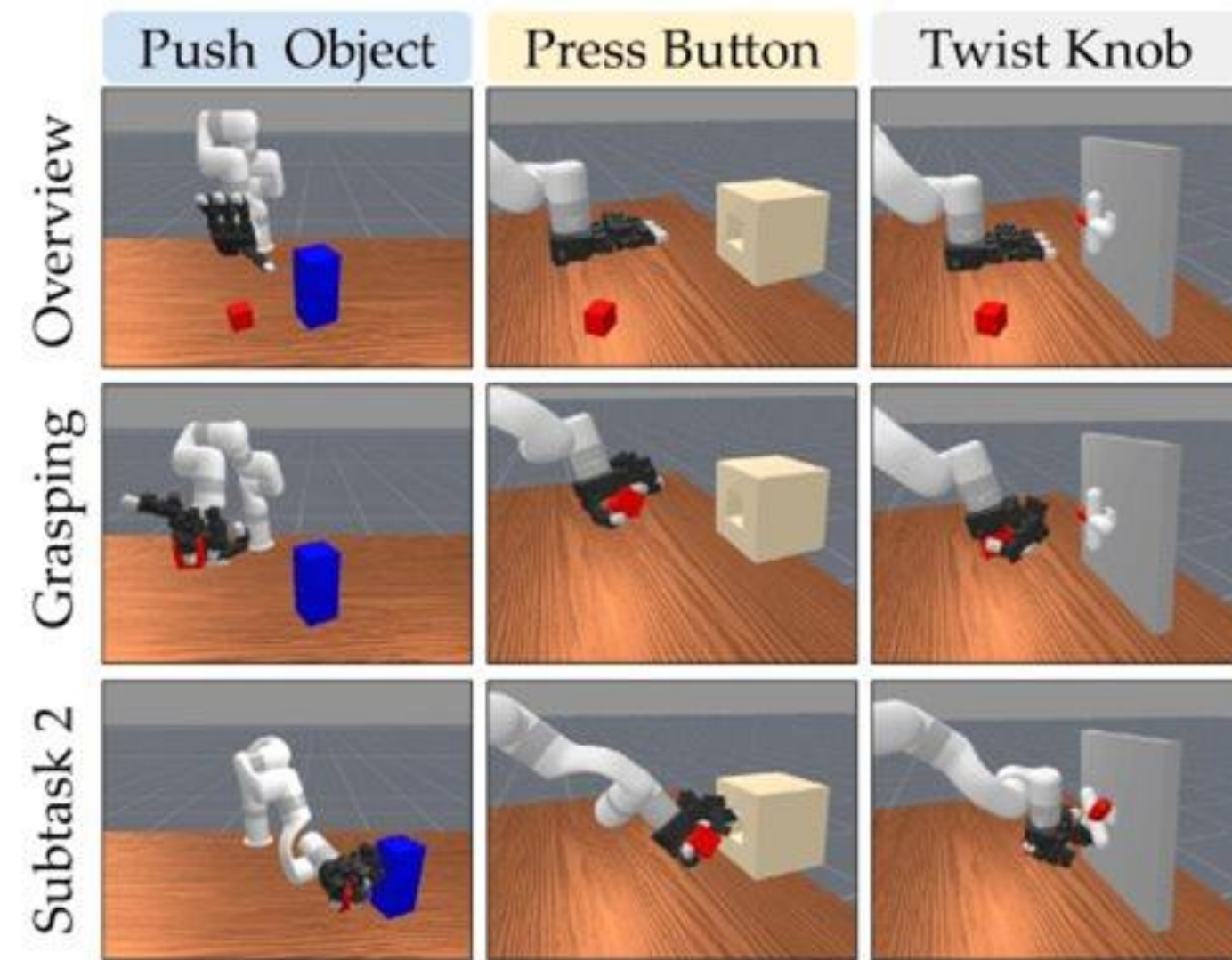


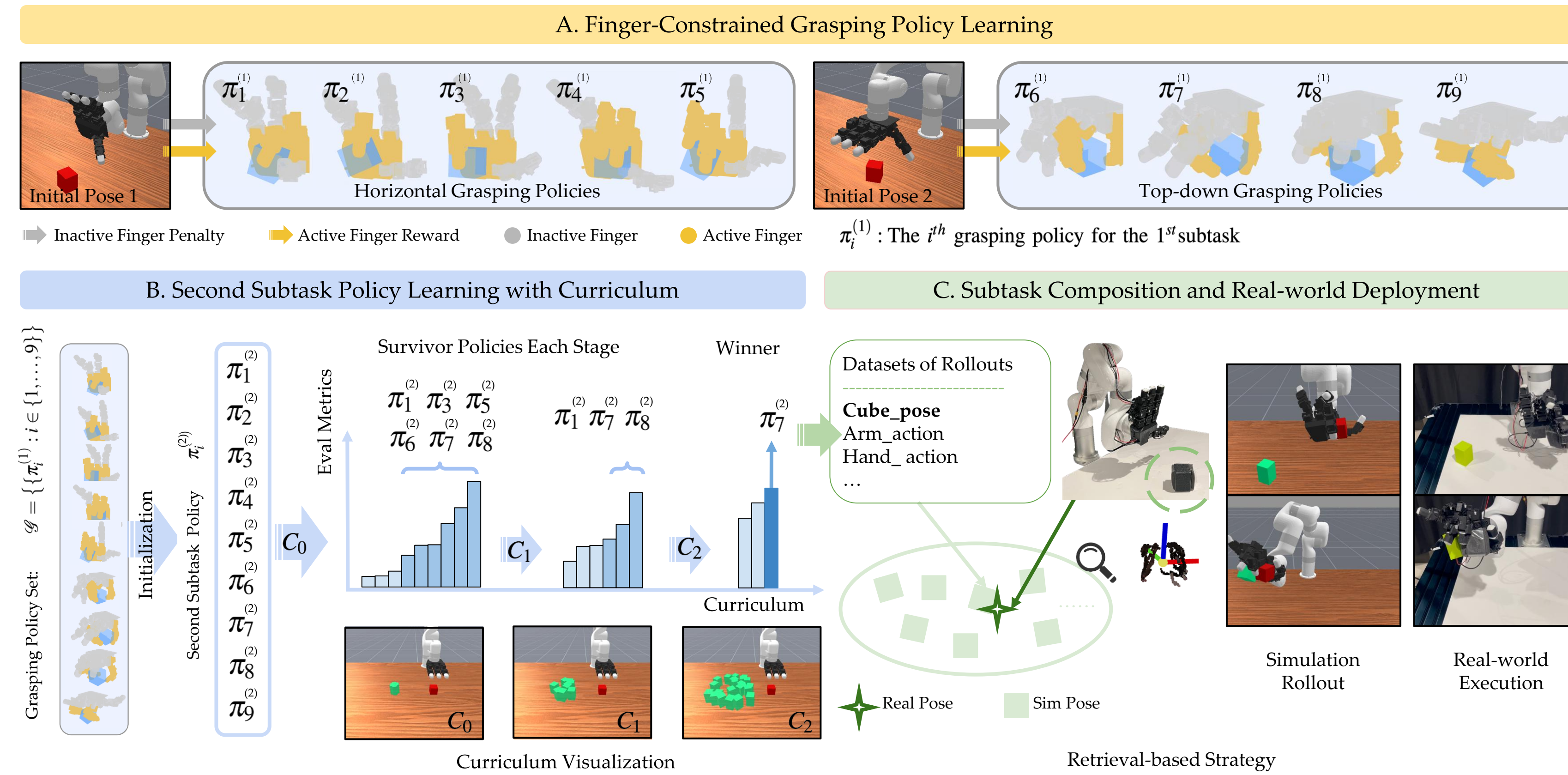


## Motivation

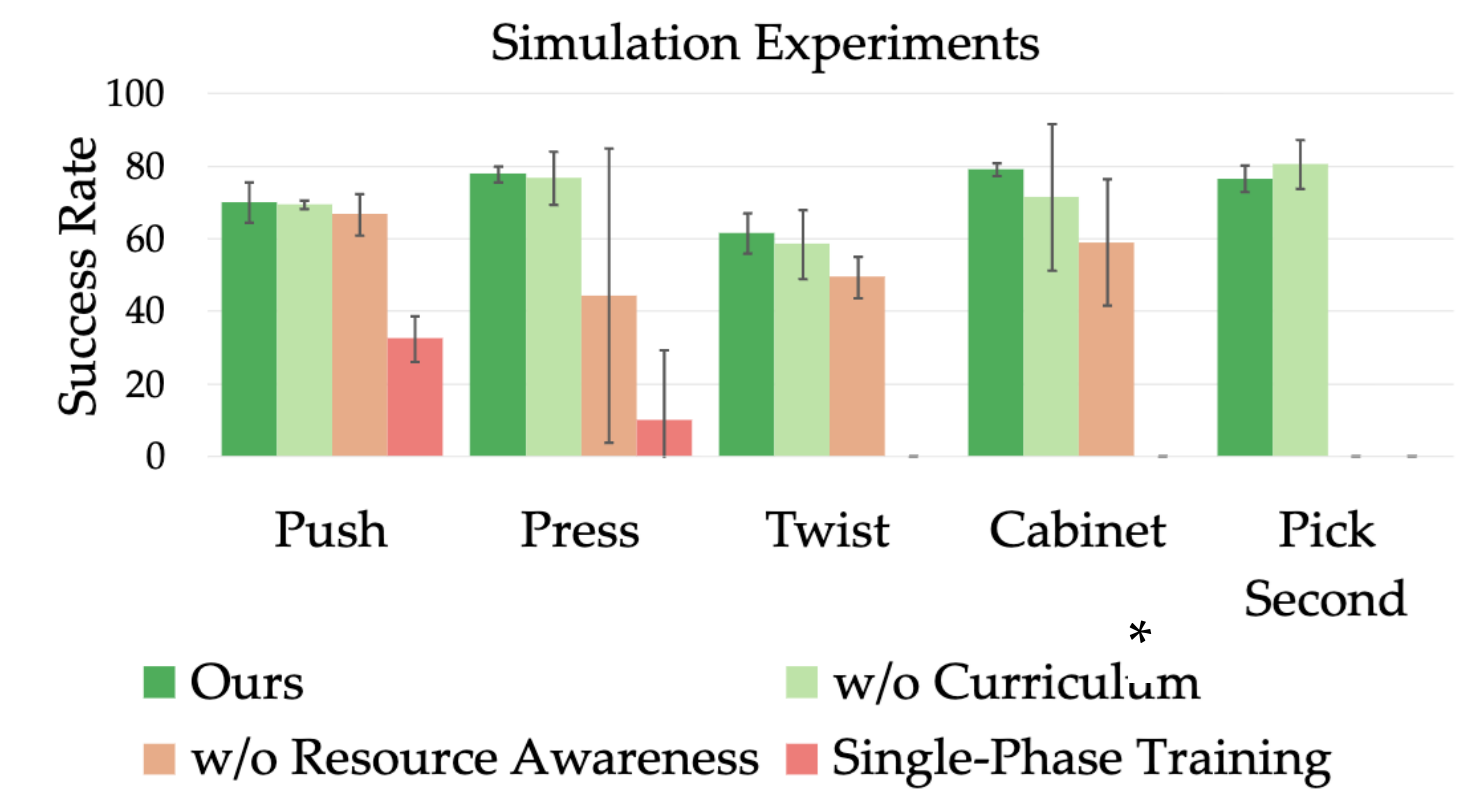
- **Multifunctional manipulation**, where a robot manipulates multiple objects simultaneously (holding a fork + pulling a drawer), are common and useful for increasing manipulation efficiency.
- Dexterous hands, while capable of executing these types of manipulations, often struggle to do so because of the **difficulty of splitting in-hand resources** (fingers or in-hand space) **between multiple skills**.



**TLDR:** We learn resource-aware one and two-finger grasps (A), leverage the leftover fingers and in-hand space from grasping for downstream objectives (B) and then transfer to real with a trajectory retrieval strategy (C).



## Results



\* Compared to an ablation w/o Curriculum Learning, HANDFUL reduces training time by 40%

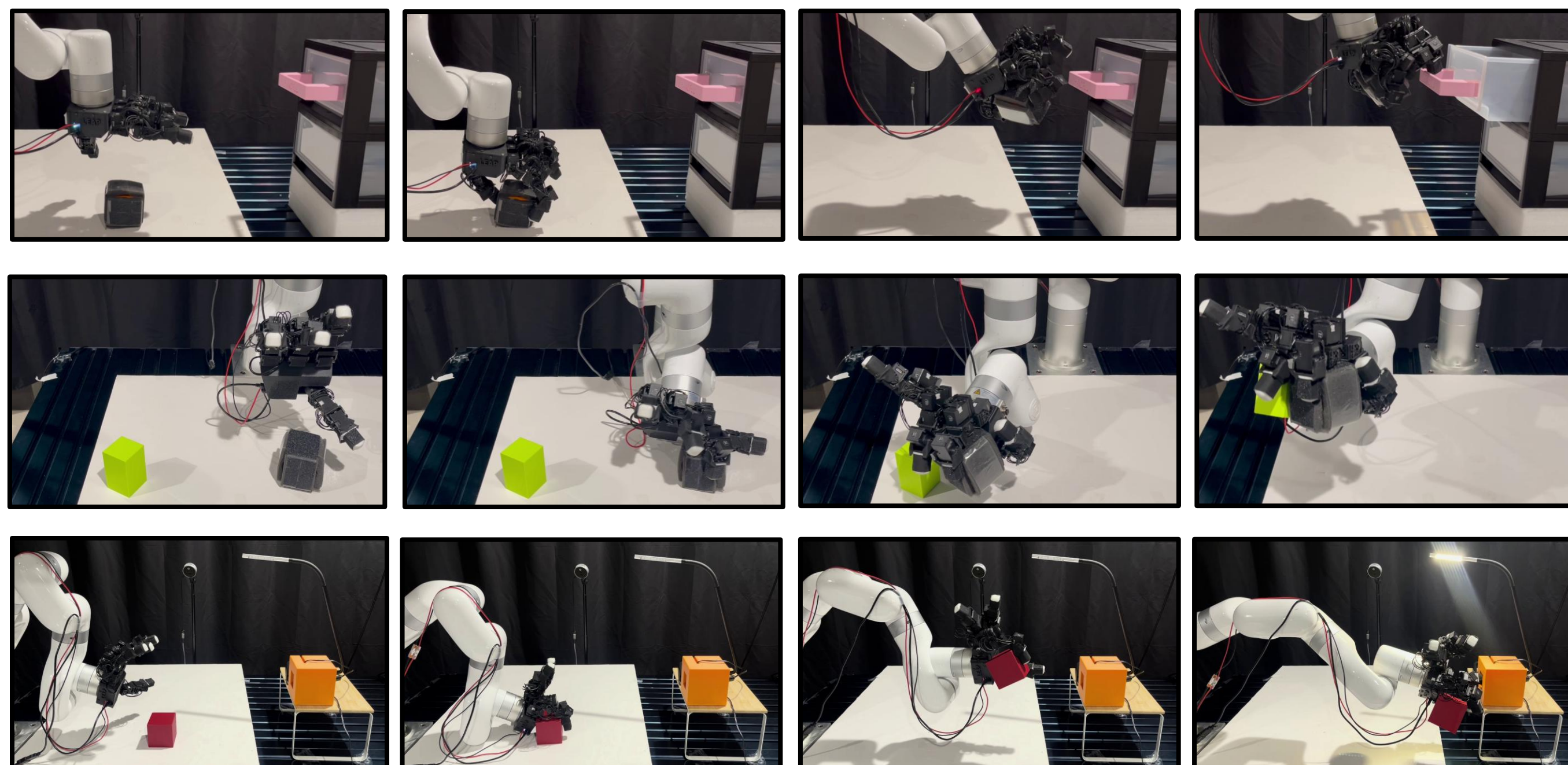
| Curriculum | $\pi_1^{(2)}$ | $\pi_2^{(2)}$ | $\pi_3^{(2)}$ | $\pi_4^{(2)}$ | $\pi_5^{(2)}$ | $\pi_6^{(2)}$ | $\pi_7^{(2)}$ | $\pi_8^{(2)}$ | $\pi_9^{(2)}$ |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $C_0$      | 51.8          | 0.0           | 42.8          | 0.0           | 0.0           | 29.4          | 36.4          | 18.4          | 1.6           |
| $C_1$      | 65.2          | 0.0           | 31.8          | 0.0           | 11.8          | 27.4          | 41.2          | 22.5          | 3.6           |
| $C_2$      | 77.8          | 0.0           | 41.8          | 0.1           | 13.2          | 39.3          | 66.0          | 52.2          | 33.3          |

Performance of different grasp types on the “Pick Second” task. Early curriculum performance accurately reflects final stage performance.

| Curriculum | $\pi_1^{(2)}$ | $\pi_2^{(2)}$ | $\pi_3^{(2)}$ | $\pi_4^{(2)}$ | $\pi_5^{(2)}$ | $\pi_6^{(2)}$ | $\pi_7^{(2)}$ | $\pi_8^{(2)}$ | $\pi_9^{(2)}$ |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $C_0$      | 3             | 3             | 3             | 3             | 3             | 3             | 3             | 3             | 3             |
| $C_1$      | 3             | 0             | 3             | 0             | 1             | 3             | 3             | 2             | 3             |
| $C_2$      | 3             | 0             | 1             | 0             | 0             | 1             | 2             | 1             | 1             |

The number of times (out of 3 seeds) each grasp type reached each stage of the curriculum for the Pick Second task. Top performing grasps perform well consistently across curriculum seeds.

## Real World Rollouts



- Zero-shot deployments of simulation policies with retrieval-based sim-to-real strategy. The strategy’s biggest challenge is maintaining resource-aware grasps during downstream tasks.

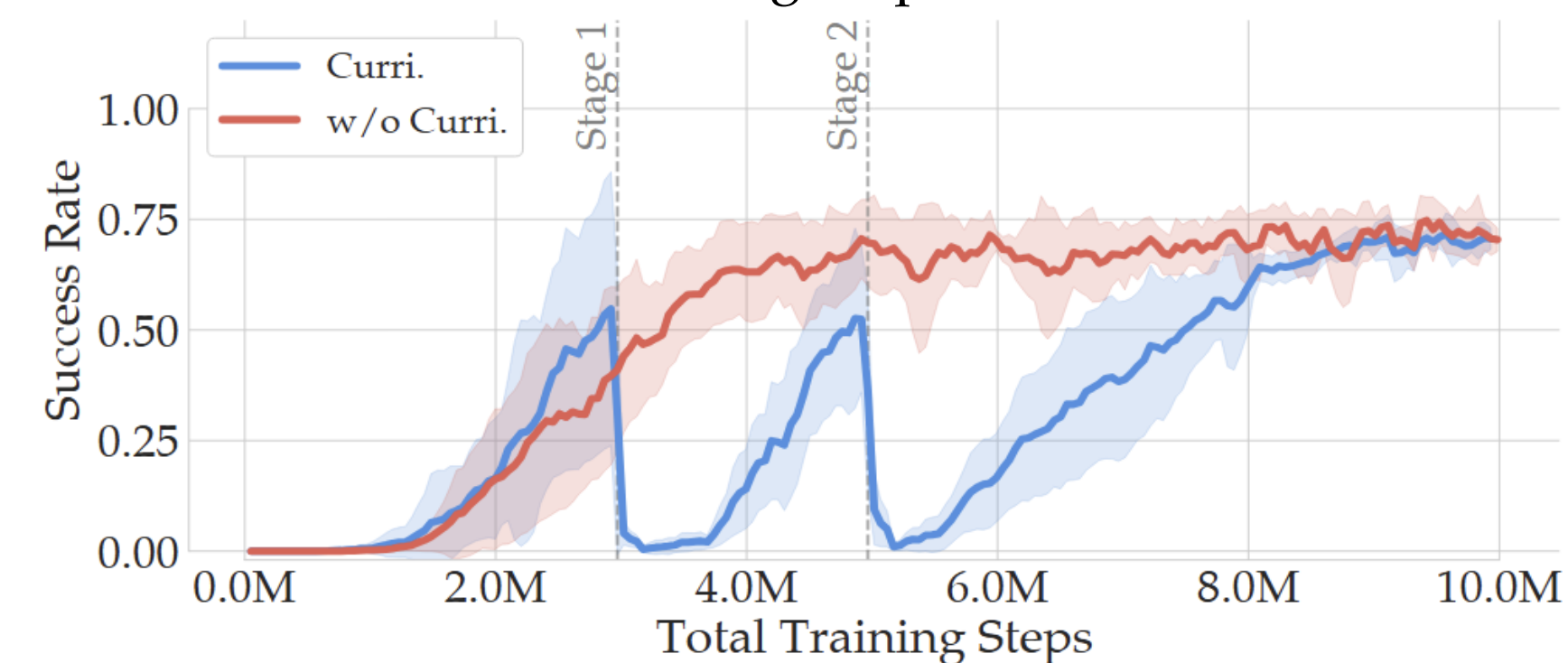
## Real World Results

| Tasks               | Push Object | Press Button | Twist Knob | Pull Drawer | Pick Second |
|---------------------|-------------|--------------|------------|-------------|-------------|
| Success (Both)      | 10          | 8            | 6          | 6           | 4           |
| Fail Subtask 1 Only | 4           | 4            | 6          | 6           | 7           |
| Fail Subtask 2 Only | 0           | 0            | 0          | 0           | 1           |
| Fail Both           | 1           | 3            | 3          | 3           | 3           |

Success and failure types (out of 15) for each task in the real world.

## Curriculum Training Curves

Success Rate vs. Training Steps for Pick Second Task



Performance with curriculum learning vs without curriculum learning over 10 million steps.

## Conclusion and Future Work

- We present HANDFUL, a pipeline for sequential, grasp-conditioned manipulation tasks.
- We hope this work inspires future investigation into multifunctional manipulation tasks with an emphasis on generating intra-skill diversity and learning methods for skill composition.

